

Here is the structured conceptual glossary of your theory (Topological Substrate Matrix - TSM), prepared on the basis of the provided notes. The conceptual framework has been synthesized to serve as a quick compendium of knowledge regarding the structure and phenomena within the model.

TSM Glossary

I. Foundations of the Medium

- **0-Particle:** The simplest, fundamental unit of the universe. It possesses an invariant rest mass (m_0) and an identical, discrete momentum of a single impact (p_0).
- **Oscillation Sphere:** A local spatial cell assigned to a 0-particle. A change in its volume (due to spatial distortion) directly determines the particle's local oscillation frequency (f_{osc}).
- **0-Matrix:** A global continuum formed by the arrangement of oscillation spheres. It creates a structured, like crystalline network (locking/restriction phenomenon) through which only wave deformations propagate.
- **Zero State:** The relaxed, perfectly symmetric 0-Matrix. All oscillation spheres have an identical volume, and the impact frequency of 0-particles is uniform. The physical equivalent of "empty space."

II. Matter, Gravity, and Energy

- **Elementary Particle (Topological Knot):** A durable, stable geometric defect (knot) bent into the fourth spatial dimension (x^4). It is not constructed from a different material but constitutes a trapped distortion of the 0-Matrix itself.
- **Mass (Gravity / Inertia):** The kinetic resistance of the 0-Matrix to distortion. It is the result of "projecting" (the shadow of) a 4D knot onto the 3-brane. This projection compresses local oscillation spheres, causing a sharp spike in the impact frequency (f_{osc}) of 0-particles.
- **Photon:** A quantized, transverse elastic wave (shear wave) in the 0-Matrix. It does not possess a permanent topological knot in 4D (no rest mass). It represents a directional transfer of excess momentum (p_0) between spheres.
- **Speed of Light (c):** The physical, kinematic strength limit of the 0-Matrix. The maximum velocity at which a mechanical shear wave can propagate through undisturbed space.

III. Electromagnetism

- **Electric Charge:** The chirality (handedness) of a topological knot projected onto the planes of the 3-brane. A knot can be right-handed or left-handed. This generates a shear (torsional) stress in the 0-Matrix. The charge value is quantized because the twist flux of a closed loop around the core must be a multiple of 2π .
- **Electrostatic Interaction:** The result of torsional stress interference. Like charges (matching chirality) collide, increasing local shear pressure (mechanical repulsion). Opposite charges compensate for each other, lowering the Matrix pressure (kinetic attraction).
- **Magnetic Field:** The dynamic vorticity of shear stresses. A vibrating "wake" (turbulence) left in the structured 0-Matrix by a moving, twisted topological knot.
- **Absence of Magnetic Monopoles:** A consequence of the medium's continuity. A free starting point of vorticity would require a physical tearing of the crystalline lattice of the 0-Matrix, which would demand infinite energy.

IV. Time as an Emergent Parameter

- **Global Statistical Clock (τ):** Time is not an inherent dimension. It is an emergent measure of the rate of events. The parameter τ is the normalized envelope of the total number of elementary impacts (N_{total}) across the entire 0-Matrix.
- **Local Time (t):** Measured by local oscillators. It depends on the local frequency f_{lok} , which varies under the influence of local topological stresses.
- **Time Dilation:** A purely mechanical phenomenon. The change in the rate of processes relative to an external observer results from the difference in the local impact frequency of oscillation spheres within the deformed geometry, rather than from the curvature of a time axis.
- **Arrow of Time:** A consequence of thermodynamics and stress dissipation within the medium. Localized, low-entropy deformations naturally tend toward relaxation.